AIRWINDOWS / VISUAL INTERFACE

CSTRIP2

VISUAL

User manual · English



The original CStrip2 sound, with a visual interface to understand and shape it.

DSP ORIGINAL: CHRIS JOHNSON / AIRWINDOWS
INDEPENDENT GUI & PORT: AMR

ENGLISH EDITION

01 · GET STARTED

One tool, four stages

CStrip2 Visual brings together filtering, compression, EQ, and a characterful output stage. Use each section on its own or combine them to shape your sound.

The signal path

01 CAPACITOR

02 BUTTERCOMP

03 EQ

04 OUTPUT

Order matters: **Capacitor** filters before **ButterComp**; the **EQ** comes next, followed by **Output**. Changes early in the chain can affect the stages that follow.

Your first test, in two minutes

01 Start with a familiar passage. Listen before touching any controls.

02 Try Capacitor HI-PASS or LO-PASS; start with a small cut.

03 Raise COMPRES until you see GR moving; listen to the result.

04 Make a small Bass, Mid, or Treble adjustment and watch Tone Curve.

05 Explore Output from 0.33; compare at a similar listening level.

> **GOLDEN RULE**
>
> You do not have to use every module. One small improvement is worth more than five unnecessary adjustments.

Where to find things

Pages 4–5 · Capacitor and ButterComp / **Pages 6–9** · EQ, crossover points, and colour trick
Page 10 · Output / **Pages 11–13** · Analyzer and comparisons
Page 14 · Presets and A/B / **Pages 15–16** · Quick reference

02 · ORIENTATION

Meet the interface

Presets and A/B comparison sit at the top; the analyzer and Tone Curve are in the centre; the processing modules are along the bottom.



Full interface when CStrip2 Visual first loads, at its default settings.

What you hear

Capacitor, ButterComp, EQ, and Output are the four audio-processing stages.

What you see

IN, OUT, GR, Spectrum Analyzer, and Tone Curve help you understand the changes. The displays do not alter audio.

The **IN** and **OUT** meters show signal levels entering and leaving the plugin. **GR** shows how much ButterComp is reducing gain. These are different measurements.

03 · CAPACITOR

Filtering the extremes

Capacitor lets you roll off the lowest and highest frequencies before the signal reaches the compressor. It is useful for cleaning up a sound without another plugin.



HI-PASS progressively reduces deep bass. It can remove rumble and unwanted sub-bass.

LO-PASS rolls off the highest frequencies. It can soften a harsh sound or unwanted high-frequency noise.



Tone Curve shows how both filters roll off the frequency extremes.

HOW TO BEGIN

Start with both filters open (**HiCap = 0.00** and **LowCap = 1.00**), then close them gradually. Stop when the unwanted frequencies are gone without losing body or clarity.

Its character: Capacitor combines several filtering stages in an unusual way, which can give it a different feel from simpler filters. This does not mean it models individual physical components.

04 · BUTTERCOMP

Compression you can hear

ButterComp helps control level variations and peaks. Its character depends on how much it compresses and how quickly it responds.



COMPRES sets the amount of compression.

SPEED / CompSpd controls the response speed. Even at similar gain reduction, different speeds can change the feel of attacks and movement.

Watch GR, but trust your ears

GR means **gain reduction**. At 0 dB, there is no gain reduction at that instant; larger reductions mean ButterComp is working harder. IN and OUT show signal level, not compression.



Heavy compression: GR shows substantial gain reduction.



Moderate compression: a gentler response.

A TRICK FOR LEARNING BUTTERCOMP

Start with fairly pronounced compression and a fast response, without necessarily maxing out the controls. Gradually reduce **SPEED** and listen to the changes in attack and movement. Then lower **COMPRES** until the effect feels musical. Compare at a similar loudness.

05 · EQUALIZER

Bass, Mid and Treble

The EQ shapes low-end weight, midrange presence, and high-end brightness. Each band can boost or cut from −12 to +12 dB.



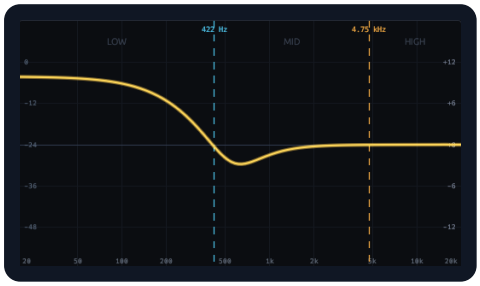
Bass · low-end weight and body.

Mid · presence and character for voices and instruments.

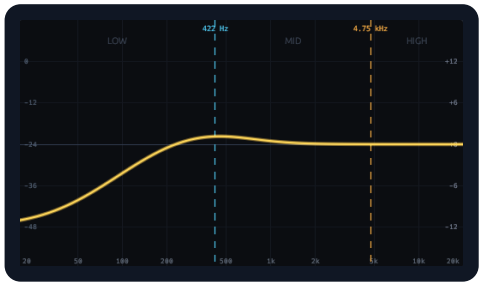
Treble · brightness and detail.

The bands interact: they are not three independent parametric EQs.

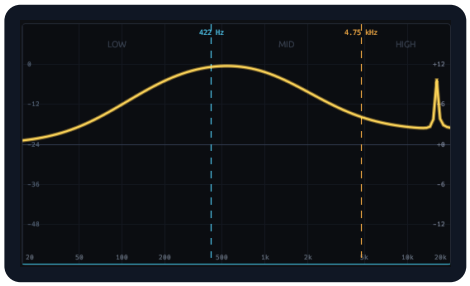
How each band responds



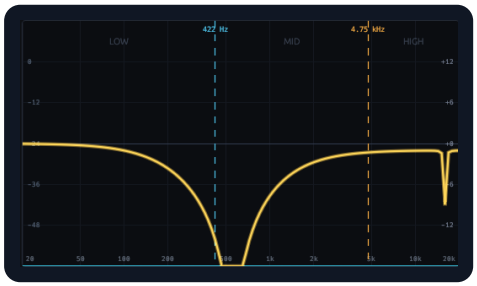
Bass +12 dB · low boost



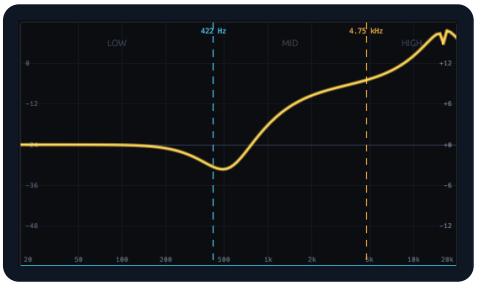
Bass −12 dB · low cut



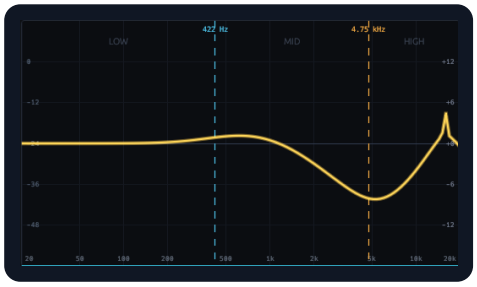
Mid +12 dB · mid boost



Mid −12 dB · mid cut



Treble +12 dB · high boost



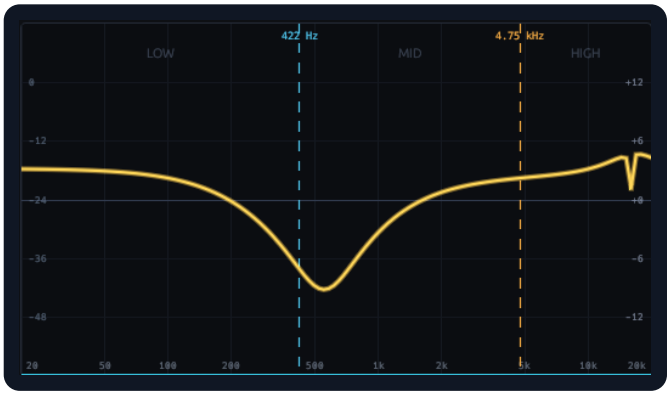
Treble −12 dB · high cut

These extreme curves help you recognise each band's shape. In a mix, smaller adjustments are often more useful.

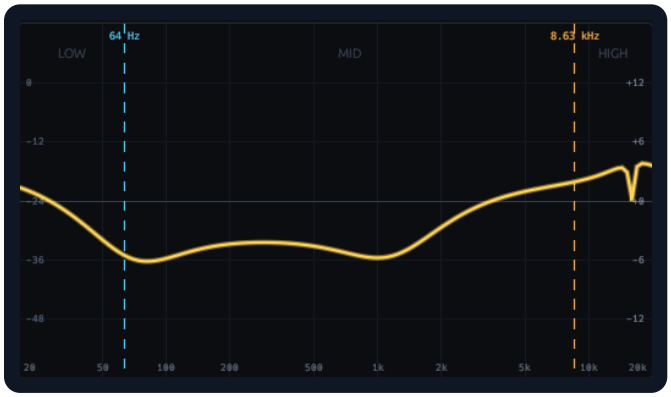
06 · EQUALIZER

Where each band works

Bass Frq and Treb Frq set the approximate crossover areas between lows, mids, and highs. Vertical dashed lines on the analyzer mark their positions.



Same EQ · original crossover settings.



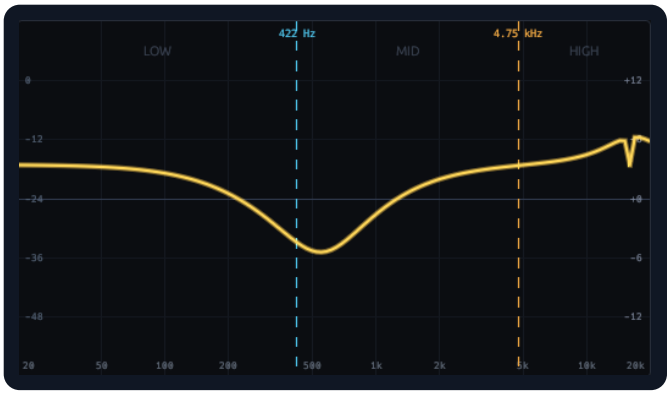
Same EQ · crossover points moved.

Compare the images: Bass, Mid, and Treble have exactly the same values in both. Only **Bass Frq and Treb Frq** change, yet the shape of the EQ curve changes considerably.

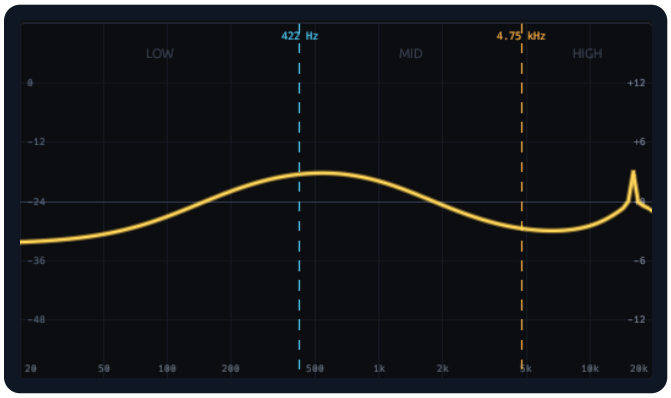
BASS FRQ roughly marks the transition from lows to mids; **TREB FRQ** marks where the treble region begins. These transitions are gradual: the dashed lines do not indicate hard cuts.

> **AN IMPORTANT DETAIL**
>
> With **Bass, Mid, and Treble** at **0 dB**, moving Bass Frq or Treb Frq does not change the EQ sound. Once you have boosted or cut a band, moving these points can change the result substantially without altering band gains.

Two different EQ shapes



An example using several EQ adjustments.

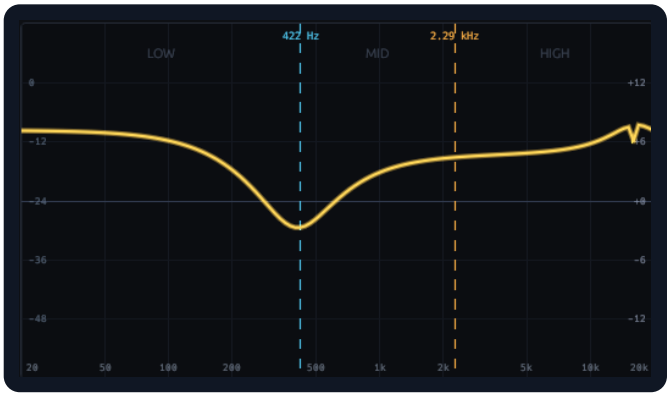


Another tonal balance using the same three bands.

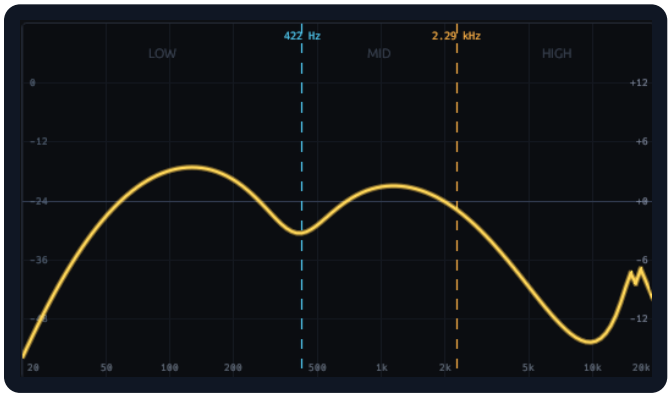
07 · TONE

EQ and Capacitor, together

Tone Curve shows the combined tonal response of Capacitor and EQ. Adjusting the filters changes the curve even if the three EQ gains stay the same.



EQ only: boosts and cuts from the three bands.



EQ + Capacitor: the frequency extremes are also reduced.

Why order matters

Capacitor operates **before ButterComp and the EQ**. Removing unwanted low end first also changes what drives the compressor. You can then shape the tone with Bass, Mid, and Treble.

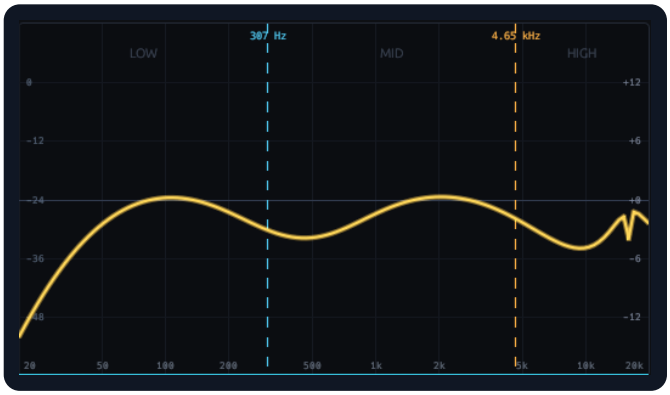
> **WHAT THE CURVE DOES NOT SHOW**
>
> Tone Curve shows the calculated frequency response of Capacitor and EQ. It **does not display ButterComp's dynamic compression** or measure harmonic saturation. Use GR for compression, and your ears for character.

Try it on a familiar track: adjust Capacitor first, then EQ, and compare the settings while listening in the mix.

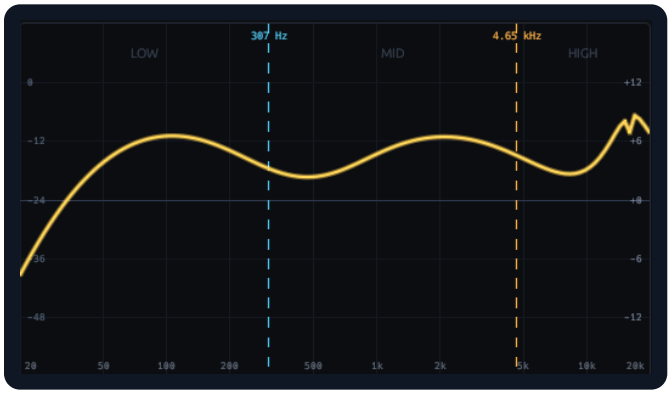
08 · EQ TRICK

Add colour, keep your EQ shape

Beyond boosting and cutting, CStrip2's EQ can add its own saturation. Each band is processed separately, so boosting all three is not the same as adding perfectly clean gain.



Original EQ: each band has its own setting.



With Shift: all three bands rise together, keeping their offsets.

What happens under the hood

Unlike a purely linear equalizer, boosts can add harmonic saturation, imparting density and making those frequencies seem to move forward into the foreground. Cuts, on the other hand, produce the opposite effect, reducing that saturation and making those frequencies seem to recede into a deeper plane in the mix.

Thus, the equalizer not only alters tonal balance: it also allows shaping the sense of presence and depth, much like a multiband saturator. (Note that desaturation does not remove previously recorded distortion.)

How to use the trick

First set Bass, Mid, and Treble the way you like. Then hold **Shift** and drag any band: all three move together, preserving their relative offsets. **You do not need to start from a flat EQ.** Stop when any band reaches −12 or +12 dB to preserve those offsets.

SUBTLE DOES NOT MEAN INAUDIBLE

With modest boosts, the saturation is usually restrained. Large boosts can make it more obvious and drive later stages harder. Adjust Output if needed and compare at a similar loudness.

OTHER EQ GESTURES

Dragging a band adjusts only that band. **Shift + drag** moves all three together. **Double-click** resets one band; **Shift + double-click** resets all three.

09 · OUTPUT

Character at the output

Output is more than a volume control. Its continuous range passes through three regions: Clean, Spiral, and Density.



CLEAN
From 0.00 to about 0.33: level adjustment with the cleanest behaviour.

SPIRAL
From about 0.33 to 0.67: increasing saturation.

DENSITY
From about 0.67 to 1.00: a stronger, denser character.

> **ONE CONTINUOUS CONTROL, THREE REGIONS**
>
> These are not three switches or selectable modes; you move through them with one control. **0.33** is the default Output setting, with no automatic loudness matching.

Try it yourself

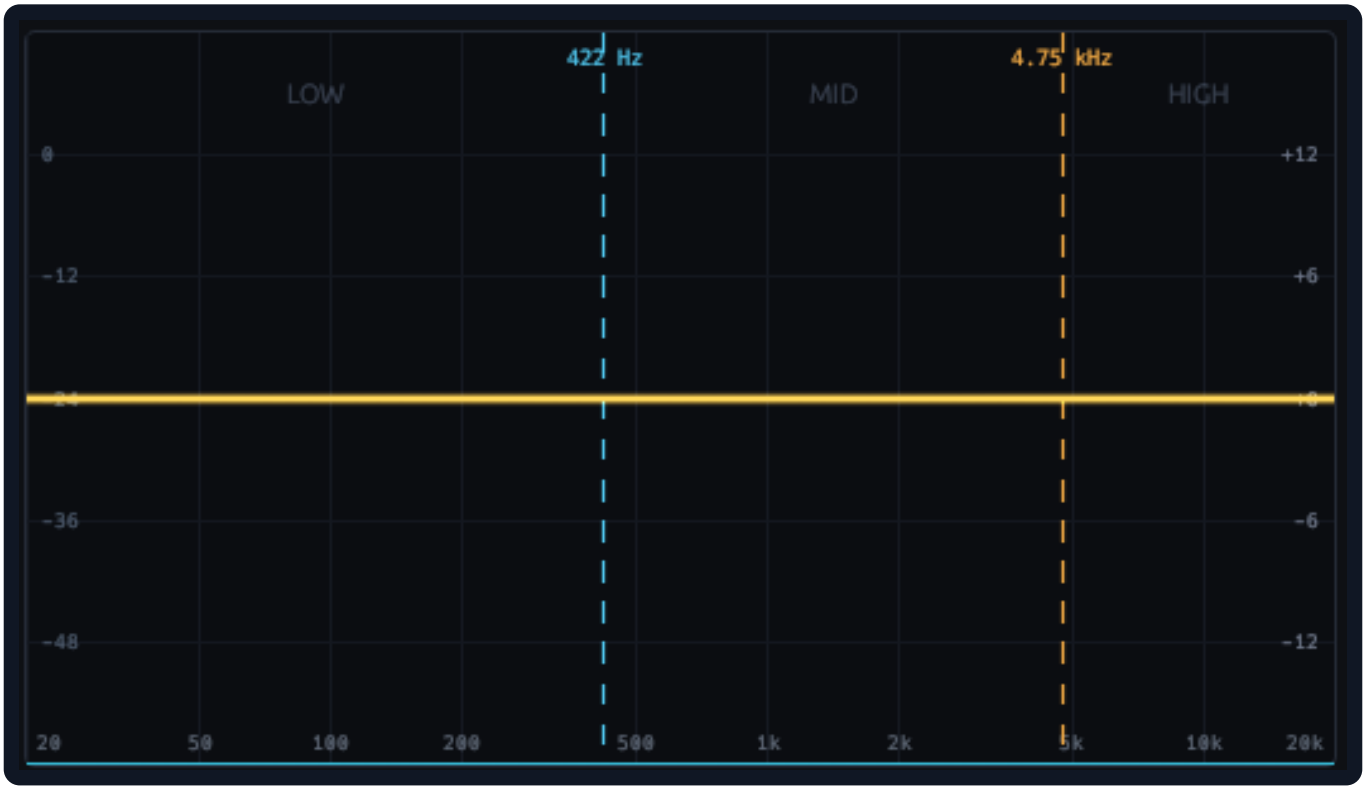
Listen to a track with Output at 0.33. Slowly move towards Spiral and Density, noting changes in tone and peaks. Then compare at a similar perceived level. Louder is not necessarily better.

Output saturation acts at the end, on the recombined signal. It is different from the colour introduced independently within the EQ bands.

10 · ANALYZER

The graph and its scales

The same display contains two different things: the spectrum of the playing audio and Tone Curve, calculated from Capacitor and EQ.



Analyzer at rest, showing its axes, markings, and scales.

EQ dB

Right-hand vertical scale: boosts and cuts in **Tone Curve**.

Signal dB

Left-hand vertical scale: signal level in each frequency region.

Frequency (Hz)

Bottom horizontal scale: low to high frequencies, spaced logarithmically.

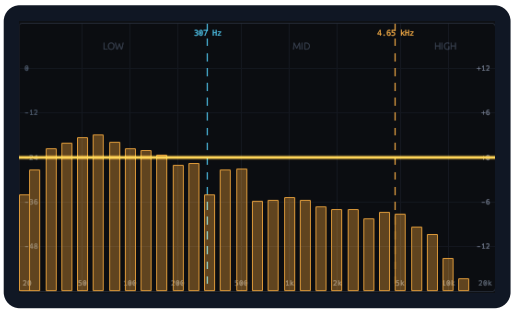
The **dashed vertical lines** show Bass Frq and Treb Frq. They move as you adjust those settings.

> **READ THE DISPLAY, TRUST YOUR EARS**
>
> Tone Curve does not show ButterComp compression or measure harmonic distortion. The analyzer is useful, but your ears make the final decision.

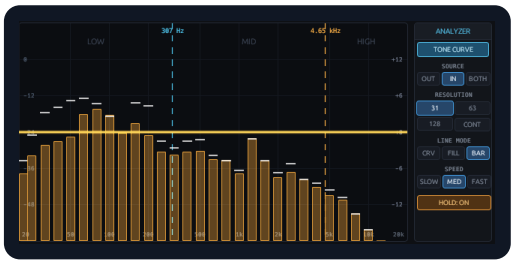
11 · ANALYZER

Choose your display

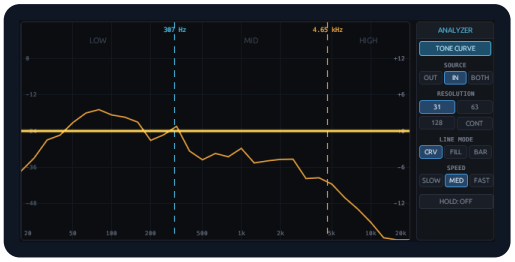
Choose the spectrum style you find easiest to read. These settings change the display, not the sound.



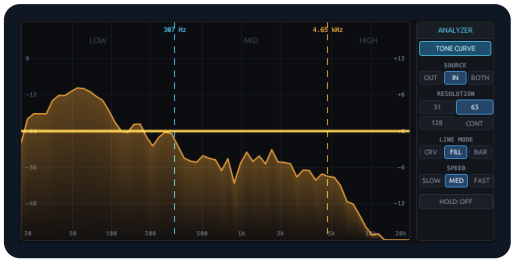
BARS · spectrum bars



BARS + HOLD · retained peaks



CURVE · line display



FILL · filled spectrum

Analyzer controls

SOURCE selects IN (orange), OUT (blue), or BOTH.

RESOLUTION offers 31, 63, 128, and CONT.

SPEED changes the visual response: SLOW, MED, or FAST.

HOLD works in all display modes.

TONE CURVE shows or hides the Capacitor + EQ response.



12 · REAL-WORLD EXAMPLE

Comparing input and output

SOURCE: BOTH shows what enters and leaves CStrip2 at the same time. This helps you spot broad spectral changes and level differences.



A running mix: IN is orange, OUT is blue, and Tone Curve is a tonal reference. Capacitor, ButterComp, EQ, and Output have all been adjusted.

A practical use for BOTH

01 Play a consistent passage with SOURCE set to BOTH.

02 Adjust Capacitor and EQ while comparing the input and output spectra.

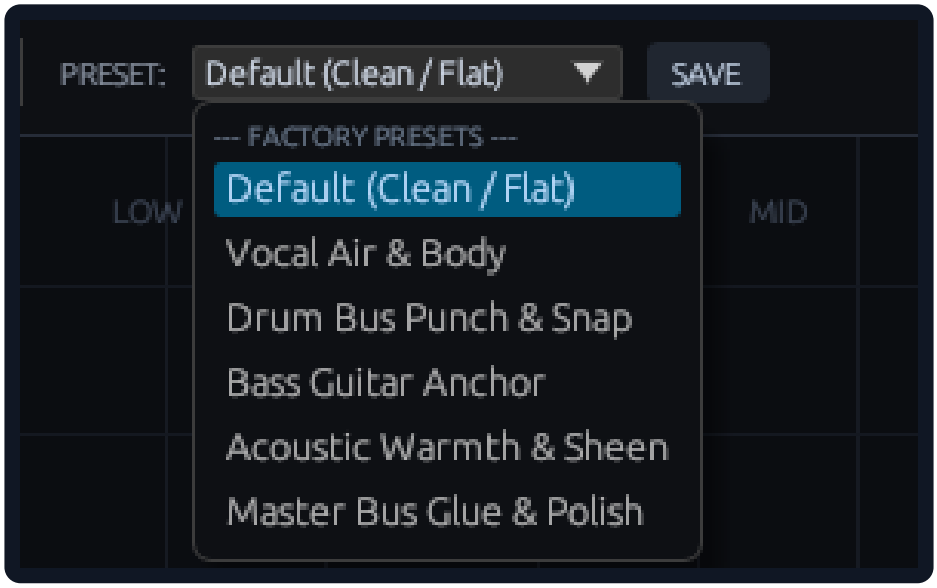
03 If you add compression or adjust Output, keep an eye on output level to avoid being misled by loudness.

04 Use your ears: similar-looking spectra can sound different, and vice versa.

13 · DAILY WORKFLOW

Presets and A/B comparison

Presets and A/B solve different problems: recalling a saved setting versus comparing two variations while you work.



Preset bar: choose a setting and use SAVE to store a user preset.

PRESET loads a setting. **SAVE** stores your current settings so you can recall them later. There is no need to write every parameter down.



A/B controls and buttons for copying settings between the two states.

A simple A/B workflow

1. Make a setting you like in A. **2.** Copy it to B. **3.** Adjust B. **4.** Switch between A and B while listening to the same passage. Copying works both ways, so take care not to overwrite the version you meant to keep.

14 · QUICK REFERENCE

Which control should I try?

These are starting points, not universal recipes. Use them to find the right control quickly.

IF YOU WANT TO...	START WITH...
Remove deep rumble	Capacitor · HI-PASS
Soften excessive high end	Capacitor · LO-PASS
Control peaks or level changes	ButterComp · COMPRES and SPEED; watch GR
Add body, presence, or brightness	EQ · Bass, Mid, Treble
Move the EQ crossover regions	Bass Frq and Treb Frq
Add per-band colour	EQ · Shift + drag any band
Add output saturation	Output · Spiral / Density
Compare before/after spectra	SOURCE · BOTH
Compare two processing options	A/B states

THREE USEFUL HABITS

Listen in context, not just in solo. **Compare at similar loudness**. And use the analyzer to understand the sound, not to replace listening.

If something sounds unexpected

With a **flat EQ**, Bass Frq and Treb Frq have no meaningful tonal effect. If **GR does not move**, check COMPRES and the incoming signal. If **Output makes everything seem better**, check how much louder it became.

15 · REFERENCE

Ranges and formats

These are the main control ranges and their default settings.

CONTROL	RANGE	DEFAULT
HI-PASS / HiCap	0.00 – 1.00	0.00 · open
LO-PASS / LowCap	0.00 – 1.00	1.00 · open
COMPRES	0.00 – 1.00	0.00
SPEED / CompSpd	0.00 – 1.00	0.00
Bass · Mid · Treble	−12 a +12 dB	0 dB
Bass Frq	30 – 1600 Hz	422.5 Hz
Treb Frq	1 – 16 kHz	4.75 kHz
Output	0.00 – 1.00	0.33

Plugin formats

Windows

VST3 and CLAP.

macOS

VST3, CLAP, and AUv2. Universal builds for Apple Silicon and Intel.

Credits and scope

Original DSP: Chris Johnson / Airwindows.
Visual interface, port, and analyzer: AMR. Independent project; not an official Airwindows release.

> **REMEMBER**
>
> Tone Curve represents **Capacitor + EQ**. GR belongs to **ButterComp**. The Output regions are **Clean, Spiral, and Density**. EQ saturation may be subtle with moderate boosts and clearly audible with large ones. Always compare at matched loudness.